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Game Thoery Gibbons Unofficial Solution Manual

Game Theory And Applications To Economics (Brown University)

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The Unofficial Solution Manual to

A Primer in Game Theory by RA Gibbons

Unfinished Draft

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This version is an unreleased and unfinished manuscript.

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This is strictly a beta version. Two thirds of it are missing and there are errors aplenty. You have been warned.

On a more positive note, if you do find an error, please email me at navin.ksrk@gmail.com, or tell me in person.

- Navin Kumar

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Static Games of Complete Information

Answer 1.1 See text.

Answer 1.2 B is strictly dominated by T. C is now strictly dominated by R. The strategies (T,M) and (L,R) survive the iterated elimination of strictly dominated strategies. The Nash Equilibrium are (T,R) and (M,L).

Answer 1.3 For whatever value Individual 1 chooses (denoted by S_1), Individual 2's best response is $S_2 = B_2(S_1) = 1 - S_2$. Conversely, $S_1 = B_1(S_2) = 1 - S_1$. We know this because if $S_2 < 1 - S_1$, then there is money left on the table and Individual 2 could increase his or her payoff by asking for more. If, however, $S_2 > 1 - S_1$, Individual 2 earns nothing and can increase his payoff by reducing his demands sufficiently. Thus the Nash Equilibrium is $S_1 + S_2 = 1$.

Answer 1.4 The market price of the commodity is determined by the formula $P = a - Q$ in which Q is determined $Q = q_1 + ... + q_n$ The cost for an individual company is given by $C_i = c \cdot q_i$. The profit made by a single firm is

$$\pi_i = (p - c) \cdot q_i = (a - Q - c) \cdot q_i = (a - q_1^* - ... - q_n^* - c) \cdot q_i$$

Where q_j^* is the profit-maximizing quantity produced by firm j in equilibrium. This profit is maximized at

$$\begin{aligned}
&\frac{d\pi_i}{dq_i} = (a - q_1^* - ... - q_i^* - ... - q_n^* - c) - q_i^* = 0 \\
&\Rightarrow a - q_1^* - ... - 2 \cdot q_i^* - ... - q_n^* - c = 0 \\
&\Rightarrow a - c = q_1^* + ... + 2 \cdot q_i^* + ... + q_n^*
\end{aligned}$$

For all i=1, ... ,n. We *could* solve this question using matrices and Cramer's rule, but a simpler method would be to observe that since all firms are symmetric, their equilibrium quantity will be the same i.e.

$$q_1^* = q_2^* = ... = q_i^* = ... = q_n^*$$

Which means the preceding equation becomes.

$$a - c = (n + 1)q_i^* \Rightarrow q_i^* = \frac{a - c}{n + 1}$$

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A similar argument applies to all other firms.

Answer 1.5 Let q_m be the amount produced by a monopolist. Thus, if the two were colluding, they'd each produce

$$q_m^1 = q_m^2 = \frac{q_m}{2} = \frac{a-c}{4}$$

In such a scenario, the profits earned by Firm 1 (and, symmetrically, Firm 2) is

$$\begin{aligned}\pi_{mm}^1 &= (P-c)\cdot q_m^1 = (a-Q-c)\cdot q_m^1 = \left(a - \frac{a-c}{2} - c\right)\cdot q_m^1 \\ &= \frac{a-c}{2}\cdot\frac{a-c}{4} = \frac{(a-c)^2}{8} = 0.13\cdot(a-c)^2\end{aligned}$$

If both are playing the Cournot equilibrium quantity, than the profits earned by Firm 1 (and Firm 2) are: [1]

$$\begin{aligned}\pi_{cc}^1 &= (P-c)\cdot q_c^1 = (a - \frac{2(a-c)}{3} - c)\cdot q_c^1 = (\frac{a+2c}{3} - c)\cdot q_c^1 \\ &= \frac{(a-c)^2}{9} \\ &= 0.11\cdot(a-c)^2\end{aligned}$$

[1] The price is determined by

$$\begin{aligned}P &= a - Q \\ &= a - q_{cc}^1 - q_{cc}^2 \\ &= a - 2\cdot\frac{a-c}{3} \\ &= \frac{a+c}{2}\end{aligned}$$

What if one of the firms (say Firm 1) plays the Cournot quantity and the other plays the Monopoly quantity? [2] Firm 1's profits are:

$$\begin{aligned}\pi_{cm}^1 &= (P-c)\cdot q_c^1 = (\frac{5a}{12} + \frac{7c}{12} - c)\cdot\frac{a-c}{3} = \frac{5}{36}\cdot(a-c)^2 \\ &= 0.14\cdot(a-c)^2\end{aligned}$$

[2] The price is given by

$$\begin{aligned}P &= a - Q \\ &= a - q_c^1 - q_m^2 \\ &= a - \frac{a-c}{3} - \frac{a-c}{4} \\ &= a - \frac{7}{12}\cdot(a-c) \\ &= \frac{5a}{12} + \frac{7c}{12}\end{aligned}$$

And Firm 2's profits are:

$$\begin{aligned}\pi_{cm}^2 &= (P-c)\cdot q_m^2 = (\frac{5a}{12} + \frac{7c}{12} - c)\cdot\frac{a-c}{4} = \frac{5}{48}\cdot(a-c)^2 \\ &= 0.10\cdot(a-c)^2\end{aligned}$$

For notational simplicity, let

$$\alpha \equiv (a-c)^2$$

Their profits are reversed when their production is. Thus, the payoffs are:

		Player 2	
		q_m	q_c
Player 1	q_m	0.13α, 0.13α	0.10α, 0.14α
	q_c	0.14α, 0.10α	0.11α, 0.11α

As you can see, we have a classic Prisoner's Dilemma: regardless of the other firm's choice, both firm's would maximize their payoff by choosing to produce the Cournot quantity. Each firm has a strictly dominated strategy ($\frac{q_m}{2}$) and they're both worse off in equilibrium (where they make $0.11\cdot(a-c)^2$ in profits) than they would have

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been had they cooperated by producing q_m together (which would have earned them $0.13 \cdot (a-c)^2$).

Answer 1.6 Price is determined by $P = a - Q$ and Q is determined by $Q = q_1 + q_2$. Thus the profit generated for Firm 1 is given by:

$$\pi_1 = (P - c_1) \cdot q_1 = (a - Q - c_1) \cdot q_1 = (a - q_1 - q_2 - c_1) \cdot q_1$$

At the maximum level of profit,

$$\frac{d\pi_1}{dq_1} = (a - c_1 - q_1 - q_2) + q_1 \cdot (-1) = 0 \Rightarrow q_1 = \frac{a - c_1 - q_2}{2}$$

And by a similar deduction,

$$q_2 = \frac{a - c_2 - q_1}{2}$$

Plugging the above equation into it's predecessor,

$$q_1 = \frac{a - c_1 - \frac{a - c_2 - q_1}{2}}{2} = \frac{a - 2c_1 + c_2}{3}$$

And by a similar deduction,

$$q_2 = \frac{a - 2c_2 + c_1}{3}$$

Now,

$$2c_2 > a + c_1 \Rightarrow 0 > a - 2c_2 + c_1 \Rightarrow 0 > \frac{a - 2c_2 + c_1}{3} \Rightarrow 0 > q_2$$
$$\Rightarrow q_2 = 0$$

Since quantities cannot be be negative. Thus, under certain conditions, a sufficient difference in costs can drive one of the firms to shut down.

Answer 1.7 We know that,

$$q_i = \begin{cases} a - p_i & \text{if } p_i < p_j, \\ \frac{a - p_i}{2} & \text{if } p_i = p_j, \\ 0 & \text{if } p_i > p_j. \end{cases}$$

We must now prove $p_i = p_j = c$ is the Nash Equilibrium of this game. To this end, let's consider the alternatives exhaustively.

If $p_i > p_j = c$, $q_i = 0$ and $\pi_j = 0$. In this scenario, Firm j can increase profits by charging $p_j + \varepsilon$ where $\varepsilon > 0$ and $\varepsilon < p_i - p_j$, raising profits. Thus this scenario is not a Nash Equilibrium.

If $p_i > p_j > c$, $q_i = 0$ and $\pi_i = 0$ and Firm i can make positive profits by charging $p_j - \varepsilon > c$. Thus, this cannot be a Nash Equilibrium.

If $p_i = p_j > c$, $\pi_i = (p_i - c) \cdot \frac{a - p_i}{2}$. Firm i can increase profits by charging $p_i - \varepsilon$ such that $p_j > p_i - \varepsilon > 0$, grab the entire market and a larger profit, provided

$$(p_i - \varepsilon - c) \cdot (a - p_i - \varepsilon) > (p_i - c) \cdot \frac{(a - p_i)}{2}$$

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Therefore, this is not a Nash Equilibrium

If $p_i = p_j = c$, then $\pi_i = \pi_j = 0$. Neither firm has any reason to deviate; if FFirm i were to reduce p_i, π_i would become negative. If Firm i were to raise p_i, $q_i = \pi_i = 0$ and he would be no better off. Thus Firm i (and, symmetrically, Firm j) have no incentive to deviate, making this a Nash Equilibrium.

Answer 1.8 The share of votes received by a candidate is given by

$$S_i = \begin{cases} x_i + \frac{1}{2} \cdot (x_j - x_i) & \text{if } x_i < x_j, \\ \frac{1}{2} & \text{if } x_i = x_j, \\ (1 - x_i) + \frac{1}{2} \cdot (x_i - x_j) & \text{if } x_i > x_j. \end{cases}$$

We aim to prove the Nash Equilibrium is $(\frac{1}{2}, \frac{1}{2})$. Let us exhaustively consider the alternatives.

Suppose $x_i = \frac{1}{2}$ and $x_j > \frac{1}{2}$ i.e. One candidate is a centrist while the other (Candidate j) isn't. In such a case, Candidate j can increase his share of the vote by moving to the left i.e. reducing x_j. If $x_j < \frac{1}{2}$, Candidate j can increase his share of the vote by moving to the right.

Suppose $x_i > x_j > \frac{1}{2}$; Candidate i can gain a larger share by moving to a point $x_j > x_i > \frac{1}{2}$. Thus this is not a Nash Equilibrium. Suppose $x_i = x_j = \frac{1}{2}$; the share of i and j are $\frac{1}{2}$. If Candidate i were to deviate to a point (say) $x_i > \frac{1}{2}$, his share of the vote would decline. Thus $(\frac{1}{2}, \frac{1}{2})$ is a unique Nash Equilibrium. This is the famous Median Voter Theorm, used extensively in the study of politics. It explains why, for example, presidential candidates in the US veer sharply to the center as election day approaches.

Answer 1.9 See text.

Answer 1.10 (a) Prisoner's Dilemma

		Player 2	
		$(p)Mum$	$(1-p)Fink$
Player 1	$(q)Mum$	$-1, -1$	$-9, 0$
	$(1-q)Fink$	$0, -9$	$-6, -6$

Prisoner's Dilemma

In a mixed strategy equilibrium, Player 1 would choose q such that Player 2 would be indifferent between Mum and Fink. The payoff from playing Mum and Fink must be equal. i.e.

$$-1 \cdot q + -9 \cdot (1 - q) = 0 \cdot q + -6 \cdot (1 - q) \Rightarrow q = -3.5$$

This is impossible.

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(b)

		Player 2		
		$Left(q_0)$	$Middle(q_1)$	$Right(1-q_0-q_1)$
Player 1	$Up(p)$	1, 0	1, 2	0, 1
	$Down(1-p)$	0, 3	0,1	2, 0

Figure 1.1.1.

Here, Player 1 must set p so that Player 2 is indifferent between Left, Middle and Right. The payoffs from Left and Middle, for example, have to be equal. i.e.

$$p \cdot 0 + (1-p) \cdot 3 = p \cdot 2 + (1-p) \cdot 1$$
$$\Rightarrow p = 0.5$$

Similarly, the payoffs from Middle and Right have to be equal

$$2 \cdot p + 1 \cdot (1-p) = 1 \cdot p + 0 \cdot (1-p)$$
$$\Rightarrow p = -0.5$$

Which, besides contradicting the previous result, is quite impossible.

(c)

		Player 2		
		$L(q_0)$	$C(q_1)$	$R(1-q_0-q_1)$
Player 1	$T(p_0)$	0, 5	4, 0	5, 3
	$M(p_1)$	4, 0	0, 4	5, 3
	$B(1-p_0-p_1)$	3, 5	3, 5	6, 6

Figure 1.1.4.

In a mixed equilibrium, Player 1 sets p_0 and p_1 so that Player 2 would be indifferent between L, C and R. The payoffs to L and C must, for example, be equal i.e.

$$4 \cdot p_0 + 0 \cdot p_1 + 5 \cdot (1-p_0-p_1) = 0 \cdot p_0 + 4 \cdot p_1 + 5 \cdot (1-p_0-p_1)$$
$$\Rightarrow p_0 = p_1$$

Similarly,

$$0 \cdot p_0 + 4 \cdot p_1 + 5 \cdot (1-p_0-p_1) = 3 \cdot p_0 + 3 \cdot p_1 + 6 \cdot (1-p_0-p_1)$$
$$\Rightarrow p_1 = 2.5 - p_0$$

Which violates $p_0 = p_1$.

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Answer 1.11 This game can be written as

		Player 2		
		$L(q_0)$	$C(q_1)$	$R(1-q_0-q_1)$
	$T(p_0)$	2, 0	1, 1	4, 2
Player 1	$M(p_1)$	3, 4	1, 2	2, 3
	$B(1-p_0-p_1)$	1, 3	0, 2	3, 0

In a mixed Nash Equilibrium, Player 1 sets p_0 and p_1 so that the expected payoffs from L and C are the same. i.e.

$$E_2(L) = E_2(C)$$
$$\Rightarrow 0 \cdot p_0 + 4 \cdot p_1 + 3 \cdot (1 - p_0 - p_1) = 1 \cdot p_0 + 2 \cdot p_1 + 2 \cdot (1 - p_0 - p_1)$$
$$\Rightarrow p_1 = 2 \cdot p_0 - 1$$

Similarly,

$$E_2(C) = E_2(R)$$
$$\Rightarrow 1 \cdot p_0 + 2 \cdot p_1 + 2 \cdot (1 - p_0 - p_1) = 2 \cdot p_0 + 3 \cdot p_1 + 0 \cdot (1 - p_0 - p_1)$$
$$\Rightarrow p_1 = \frac{2}{3} - p_0$$

Combining these,

$$2 \cdot p_0 - 1 = \frac{2}{3} - p_0 \Rightarrow p_0 = \frac{5}{9}$$
$$\therefore p_1 = 2 \cdot p_0 - 1 = 2 \cdot \frac{5}{9} - 1 = \frac{1}{9}$$
$$\therefore 1 - p_0 - p_1 = 1 - \frac{5}{9} - \frac{1}{9} = \frac{3}{9}$$

Now we must calculate q_0 and q_1. Player 2 will set them such that

$$E_1(T) = E_1(M)$$
$$\Rightarrow 2 \cdot q_0 + 1 \cdot q_1 + 4 \cdot (1 - q_0 - q_1) = 3 \cdot q_0 + 1 \cdot q_1 + 2 \cdot (1 - q_0 - q_1)$$
$$\Rightarrow q_1 = 1 - 1.5 \cdot q_0$$

And

$$E_1(M) = E_1(B)$$
$$\Rightarrow 3 \cdot q_0 + 1 \cdot q_1 + 2 \cdot (1 - q_0 - q_1) = 1 \cdot q_0 + 0 \cdot q_1 + 3 \cdot (1 - q_0 - q_1)$$

Qed

Answer 1.12

	$(q)L_2$	$(1-q)R_2$
$(p)T_1$	2, 1	0, 2
$(1-p)B_1$	1, 2	3, 0

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Player 1 will set p such that

$$E_2(L) = E_2(R)$$

$$\Rightarrow 1 \cdot p + 2 \cdot (1-p) = 2 \cdot p + 0 \cdot (1-p)$$

$$\Rightarrow p = \frac{2}{3}$$

Player 2 will set q such that

$$E_1(T) = E_1(B)$$

$$\Rightarrow 2 \cdot q + 0 \cdot (1-q) = 1 \cdot q + 3 \cdot (1-q)$$

$$\Rightarrow q = \frac{3}{4}$$

Answer 1.13

	(q)Apply^1 to Firm 1	$(1-q)$Apply^1 to Firm 2
(p)Apply^2 to Firm 1	$\frac{1}{2}w_1$,$\frac{1}{2}w_1$	w_1,w_2
$(1-p)$Apply^2 to Firm 2	w_2,w_1	$\frac{1}{2}w_2$,$\frac{1}{2}w_2$

There are two pure strategy Nash Equilibrium (Apply to Firm 1, Apply to Firm 2) and (Apply to Firm 2, Apply to Firm 1). In a mixed-strategy equilibrium, Player 1 sets p such that Player 2 is indifferent between Applying to Firm 1 and Applying to Firm 2.

$$E_2(\text{Firm 1}) = E_2(\text{Firm 2})$$

$$\Rightarrow p \cdot \frac{1}{2}w_1 + (1-p) \cdot w_1 = p \cdot w_2 + (1-p) \cdot \frac{1}{2}w_2$$

$$\Rightarrow p = \frac{2w_1 - w_2}{w_1 + w_2}$$

Since $2w_1 > w_2$, $2w_1 - w_2$ is positive and $p > 0$. For $p < 1$ to be true, it must be the case that

$$\frac{2w_1 - w_2}{w_1 + w_2} < 1 \Rightarrow \frac{1}{2}w_1 < w_2$$

Which is true. And since the payoffs are symmetric, a similar analysis would reveal that

$$q = \frac{2w_1 - w_2}{w_1 + w_2}$$

Answer 1.14

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Dynamic Games of Complete Information

Answer 2.1 The total family income is given by

$$I_C(A) + I_P(A)$$

This is maximized at

$$\frac{d(I_C(A) + I_P(A))}{dA} = 0 \Rightarrow \frac{dI_C(A)}{dA} = -\frac{dI_P(A)}{dA}$$

The utility function of the parents is given by

$$V(I_P - B) + kU(I_C + B)$$

This is maximized at

$$k\frac{dU(I_C + B)}{dB} + \frac{dV(I_P - B)}{db} = 0$$

$$\Rightarrow k\frac{dU(I_C + B)}{d(I_C + B)} \cdot \frac{d(I_C + B)}{dB} + \frac{dV(I_P - B)}{d(I_P - B)} \cdot \frac{I_P - B}{dB} = 0$$

$$\Rightarrow kU'(I_C + B) - V'(I_P - B) = 0$$

$$\Rightarrow V'(I_P - B^*) = kU'(I_C + B^*)$$

Where B^* is the maximizing level of the bequest. We know it exists because a) there are no restrictions on B and b) V() and U() are concave and increasing

The child's utility function is given by $U(I_C(A) - B^*(A))$. This is maximized at

$$\frac{dU(I_C(A) + B^*(A))}{dA} = 0$$

$$\Rightarrow U'(I_C(A) + B^*(A)) \cdot \left(\frac{dI_C(A)}{dA} + \frac{dB^*(A)}{dA}\right) = 0$$

$$\Rightarrow \frac{dI_C(A)}{dA} + \frac{dB^*(A)}{dA} = 0 \Rightarrow I_C'(A) = -B'^*(A)$$

We now have only to prove $B'^*(A) = I_P'(A)$

$$\frac{dV(I_P(A) - B^*(A))}{dA} = 0$$

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$$\Rightarrow V'(I_P(A)^*_B(A)) \cdot \left(\frac{dI_P(A)}{dA} - \frac{dB^*(A)}{dA} \right) = 0$$

$$\Rightarrow I'_P(A) = B'^*(A)$$

Answer 2.2 The utility function of the parent is given by $V(I_P - B) + k[U_1(I_C - S) + U_2(B + S)]$. This is maximized at

$$\frac{d \cdot \{V(I_P - B) + k[U_1(I_C - S) + U_2(B + S)]\}}{dB} = 0$$

$$\Rightarrow -V' + [U'_1(-S'_B) + U'_2(S'_B + 1)] = 0 \Rightarrow V' = -kU'_1S' + U'_2S' + U'_2$$

The utility of the child is given by $U_1(I_C - S) + U_2(S + B)$. This is maximized at:

$$\frac{d[U_1(I_C - S) + U_2(S + B)]}{dS} = 0 \Rightarrow U'_1 = U'_2(1 + B')$$

Total utility is given by

$$V(I_p - B) + k(U_1(I_C - S) + U_2(B + S)) + U_1(I_C - S) + U_2(B + S)$$

$$= V(I_p - B) + (1 + k)(U_1(I_C - S) + U_2(B + S))$$

This is maximized (w.r.t S) at:

$$V' \cdot (-B'_S) + (1 + K) \cdot [U'_1(-1) + U'_2(1 + B'_S)] = 0$$

$$\Rightarrow U'_1 = U'_2(1 + B'_S) - \frac{V'B'_S}{1 + k}$$

as opposed to $U'_1 = U'_2(1 + B'_S)$, which is the equilibrium condition. Since $\frac{V'B'_S}{1+k} > 0$, the equilibrium U'_1 is 'too high' which means that S - the level of savings - must be too low (since $\frac{dU'}{dS} < 0$). It should be higher.

Answer 2.3 To be done

Answer 2.4 Let's suppose $c_2 = R - c_1$, partner 2's payoff is $V - (R - c_1)^2$. If $c_1 \geq R$, partner 2's best response is to put in 0 and pocket V. If $c_1 < R$ and partner 2 responds with some c_2 such that $c_2 < R - c_1$, his payoff is $-c_2^2$. If he puts in nothing, he will, receive a payoff of zero. There is, therefore, no reason to put in such a low amount. There is - obviously - also no reason to put in any $c_2 > R - c_1$. He will put in $R - c_1$ if it's better than putting in nothing i.e.

$$V - (R - c_1)^2 \geq 0 \Rightarrow c_1 \geq R - \sqrt{V}$$

For player 1, any $c_1 > R - \sqrt{V}$ is dominated by $c_1 = R - \sqrt{V}$. Now, player 1 will do this if the benefit exceeds the cost:

$$\delta V \geq (R - \sqrt{V})^2$$

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$$\Rightarrow \delta \geq \left(\frac{R}{\sqrt{V}} - 1\right)^2$$

If $R^2 \geq 4V$, δ would have to be greater than one, which is impossible. Therefore, if $\delta \geq \left(\frac{R}{\sqrt{V}} - 1\right)^2$ and $R^2 \geq 4V$ (i.e. the cost is not 'too high'), $c_1 = R - \sqrt{V}$ and $c_2 = \sqrt{V}$. Otherwise, $c_2 = 0$ and $c_1 = 0$.

Answer 2.5 Let the 'wage premium' be $p = w_D - w_E$, where $p \in (-\infty, \infty)$. In order to get the worker to acquire the skill, the firm has to credibly promise to promote him if he acquires the skill - and not promote him if he doesn't.

Let's say that he hasn't acquired the skill. The firm will *not* promote him *iff* the returns to the firm are such that:

$$y_{D0} - w_D \leq y_{E0} - w_E \Rightarrow y_{D0} - y_{E0} \leq w_D - w_E = p$$

If he *does* acquire the skill, the firm will promote if the returns to the firm are such that:

$$y_{DS} - w_D \geq y_{ES} - w_E \Rightarrow y_{DS} - y_{ES} \geq w_D - w_E = p$$

Thus the condition which the firm behaves as it ought to in the desired equilibrium is:

$$y_{D0} - y_{E0} \leq p \leq y_{DS} - y_{ES}$$

Given this condition, the worker will acquire the promotion *iff* he acquires the skill. He will acquire the skill *iff* the benefit outweighs the cost i.e.

$$w_D - C \geq w_E \Rightarrow w_E + p - C \geq w_E \Rightarrow p \geq c$$

That is, the premium paid by the company must cover the cost of training. The company wishes (obviously) to minimize the premium, which occurs at:

$$w_D - w_E = p = \begin{cases} C & \text{if } C \geq y_{D0} - y_{E0}, \\ y_{D0} - y_{E0} & \text{if } C < y_{D0} - y_{E0} \end{cases}$$

A final condition is that the wages must be greater than the alternative i.e. $w_E \geq 0$ and $w_D \geq 0$. The firm seeks to maximize $y_{ij} - w_i$, which happens at $w_E = 0$ and $w_D = p$.

Answer 2.6 The price of the good is determined by

$$P(Q) = a - q_1 - q_2 - q_3$$

The profit earned by a firm is given by $\pi_i = (p - c) \cdot q_i$. For Firm 2, for example

$$\pi_2 = (a - q_1^* - q_2 - q_3 - c) \cdot q_2$$

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which is maximized at

$$\frac{d\pi_2}{dq_2} = (a - q_1^* - q_2 - q_3 - c) + q_2(-1) = 0$$

$$\Rightarrow q_2 = \frac{a - q_1^* - q_3 - c}{2}$$

Symmetrically,

$$q_3 = \frac{a - q_1^* - q_2 - c}{2}$$

Putting these two together,

$$q_2 = \frac{a - q_1^* - \frac{a - q_1^* - q_2 - c}{2} - c}{2} \Rightarrow q_2 = \frac{a - c - q_1^*}{3}$$

Which, symmetrically, is equal to q_3.

$$\therefore \pi_1 = (a - q_1 - q_2 - q_3 - c) \cdot q_1 = (a - q_1 - 2 \cdot \frac{a - c - q_1}{3} - c) \cdot q_1 = \left(\frac{a - q_1 - c}{3}\right) \cdot q_1$$

This is maximized at

$$\frac{d\pi_1}{dq_1} = \frac{a - q_1 - c}{3} + \frac{-q_1}{3} = 0 \Rightarrow q_1^* = \frac{a - c}{2}$$

Plugging this into the previous equations, we get

$$q_2 = q_3 = \frac{a - c}{6}$$

Answer 2.7 The profit earned by firm i is

$$\pi_i = (p - w) \cdot L_i$$

Which is maximized at

$$\frac{d\pi_i}{dL_i} = \frac{d(p - w) \cdot L_i}{dL_i} = \frac{d(a - Q - w)L_i}{dL_i} = 0$$

$$\Rightarrow \frac{d(a - L_1 - \ldots - L_i - \ldots - L_n - w) \cdot L_i}{dL_i} = 0$$

$$\Rightarrow (a - L_1 - \ldots - L_i - \ldots - L_n - w) + L_i(-1) = 0$$

$$\Rightarrow L_1 + \ldots + 2L_i + \ldots + L_n = a - w \; \forall \; i = 1, \ldots, n$$

This generates a system of equation similar to the system in Question (1.4)

$$\begin{pmatrix} 2 & 1 & \ldots & 1 \\ 1 & 2 & \ldots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \ldots & 2 \end{pmatrix} \cdot \begin{pmatrix} L_1 \\ L_2 \\ \vdots \\ L_n \end{pmatrix} = \begin{pmatrix} a - w \\ a - w \\ \vdots \\ a - w \end{pmatrix}$$

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Which resolves to

$$L_i = \frac{a-w}{n+1}$$

Thus, total labor demand is given by

$$L = L_1 + L_2 + \ldots + L_n = \frac{a-w}{n+1} + \ldots + \frac{a-w}{n+1} = \frac{n}{n+1} \cdot (a-w)$$

The labor union aims to maximize

$$U = (w - w_a) \cdot L = (w - w_a) \cdot \frac{n}{n+1} \cdot (a-w) = \frac{n}{n+1} \cdot (aw - aw_a - w^2 + ww_a)$$

The union maximize U by setting w. This is maximized at

$$\frac{dU}{dw} = (a - 2w + w_a) \cdot \frac{n}{n+1} = 0 \Rightarrow w = \frac{a+w_a}{2}$$

The sub-game perfect equilibrium is $L_i = \frac{a-w}{n+1}$ and $w = \frac{a+w_a}{2}$. Although the wage doesn't change with n, the union's utility (U) is an increasing function of $\frac{n}{n+1}$, which is an increasing function of n. This is so because the more firms there are in the market, the greater the quantity produced: more workers are hired to produce this larger quantity, increasing employment and the utility of the labor union.

Answer 2.8

Answer 2.9 From section 2.2.C, we know that exports from country i can be driven to 0 *iff*

$$e_i^* = \frac{a - c - 2t_j}{3} = 0 \Rightarrow t_j = \frac{a-c}{2}$$

which, symmetrically, is equal to t_i. Note that in this model $c = w_i$ and we will only be using w_i from now, for simplicity. What happens to domestic sales?

$$\therefore h_i = \frac{a - w_i + t_i}{3} = \frac{a - w_i + \frac{a-w_i}{2}}{3} = \frac{a-w_i}{2}$$

Which is the monopoly amount. Now, in the monopoly-union bargaining model, the quantity produced equals the labor demanded. Thus,

$$L_i^{e=0} = \frac{a-w_i}{2}$$

The profit earned by firm i in this case is given by

$$\pi_i^{e=0} = (p_i - w_i)h_i = (a - w_i - h_i)h_i = \left(a - w_i - \left(\frac{a-w_i}{2}\right)\right)\left(\frac{a-w_i}{2}\right) = \left(\frac{a-w_i}{2}\right)^2$$

The union function's payoff is defined by

$$U^{e=0} = (w_i - w_a)L_i = (w_i - w_a)\left(\frac{a-w_i}{2}\right) = \frac{w_i a - w_a - w_i^2 + w_i w_a}{2}$$

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The union sets wages to maximize utility with the following condition:

$$\frac{dU^{e=0}}{dw_i} = \frac{a - 2w_i + w_a}{2} = 0 \Rightarrow w_i = \frac{a + w_a}{2}$$

Now, tariffs decline to zero. In this situation, $t_j = 0$ and therefore

$$h_j = h_i = \frac{a - w_i}{3}$$

and

$$e_j = e_i = \frac{a - w_i}{3}$$

Due to this, prices fall:

$$P_i = P_j = a - Q = a - h_i - e_j = a - \frac{a - w_i}{3} - \frac{a - w_i}{3} = \frac{a + 2w_i}{3}$$

So what happens to profits?

$$\pi_i^{t=0} = (p_i - w_i)h_i + (p_j - w_i)e_j = \left(\frac{a + 2w_i}{3} - w_i\right)\left(\frac{a - w_i}{3}\right) + \left(\frac{a + 2w_i}{3} - w_i\right)\left(\frac{a - w_i}{3}\right)$$

$$\Rightarrow \pi_i^{t=0} = 2\left(\frac{a - w_i}{3}\right)^2 = 2 \cdot \pi_i^{e=0}$$

Thus, profits are higher at zero tariffs. What happens to employment?

$$L_i^{t=0} = q_i = h_i + e_i = \frac{2}{3} \cdot (a - w_i) = \frac{4}{3} \cdot \left(\frac{a - w_i}{2}\right) = \frac{4}{3} \cdot L_i^{e=0}$$

Employment rises. And what happens to the wage? That depends on the payoff the union now faces:

$$U^{t=0} = (w_i - w_a) \cdot L_i = \frac{2}{3}(w_i - w_a)(a - w_i) = \frac{2}{3} \cdot (w_i a - w_a a - w_i^2 + w_a w_i)$$

This is maximized at

$$\frac{dU^{t=0}}{dw_i} = \frac{2}{3}(a - 2w_i + w_a) = 0 \Rightarrow w_i = \frac{a + w_a}{2}$$

Which is the same as before.

Answer 2.10 Note that (P_1, P_2), (R_1, R_2) and (S_1, S_2) are Nash Equilibrium.

	P_2	Q_2	R_2	S_2
P_1	2, 2	x, 0	-1, 0	0,0
Q_1	0, x	4, 4	-1, 0	0, 0
R_1	0, 0	0, 0	0, 2	0, 0
S_2	0, -1	0, -1	-1, -1	2, 0

So what are player 1's payoff from playing the strategy? Let $PO_i^j(X_1, X_2)$ denote player i's payoff in round j when player 1 plays X_1 and

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player 2 plays X_2. If player 2 doesn't deviate, he earns the sum of payoffs from two rounds of gaming:

$$PO_1^1(Q_1, Q_2) + PO_1^2(P_1, P_2) = 4 + 2 = 6$$

And if player 2 deviates from the strategy, player 1 earns:

$$PO_1^1(Q_1, P_2) + PO_1^2(S_1, S_2) = 0 + 2 = 2$$

Now, let's look at his payoff from deviating, when 2 doesn't:

$$PO_1^1(P_1, Q_2) + PO_1^2(R_1, R_2) = x + 0 = x$$

And when they both deviate:

$$PO_1^1(P_1, P_2) + PO_1^2(P_1, P_2) = 2 + 2 = 4$$

Thus, if player 2 deviates, player 1's best response is to deviate for a payoff of 4 (as opposed to the 2 he'd get when not deviating). If, however, player 2 doesn't deviate, player 1 gets a payoff of 6 when playing the strategy and x when he doesn't. Thus, he (player 1) will play the strategy $iff\ x < 6$. A symmetric argument applies to player 2.

Thus the condition for which the strategy is a sub-game perfect Nash Equilibrium is

$$4 < x < 6$$

Answer 2.11 The only pure-strategy subgame-perfect Nash Equilibrium in this game are (T, L) and (M, C).

	L	C	R
T	$\underline{3}$,$\underline{1}$	0,0	$\underline{5}$,0
M	2,1	$\underline{1}$,$\underline{2}$	3,1
B	1,2	0,1	4,$\underline{4}$

Unfortunately, the payoff (4,4) - which comes from the actions (B,R) - cannot be maintained. Player 2 would play R, if player 1 plays B - player 1 however would deviate to T to earn a payoff of 5. Consider, however, the following strategy for player 2:

- In Round 1, play R.
- In Round 2, if Round 1 was (B,R), play L. Else, play M.

Player 1's best response in Round 2 is obviously T or M, depending on what player 2 does. But what should he do in Round 1? If he plays T, his playoff is: [3]

$$PO_1^1(T, R) + PO_1^2(M, C) = 5 + 1 = 6$$

If he plays B:

$$PO_1^1(B, R) + PO_1^2(T, L) = 4 + 3 = 7$$

Thus, as long as player 2 is following the strategy given above, we can induce (B,R).

[3] If you do not understand the notation, see Answer 2.10

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To get a intuitive idea of how we constructed the strategy, note that there are two Nash equilibrium that player 2 can play in the second round: a "reward" equilibrium in which player 1 gets 3 and "punishment" equilibrium in which player 1 gets 1. By (credibly) threatening to punish player 1 in Round 2, player 2 induces "good" behavior in Round 1.

Answer 2.12 See text.

Answer 2.13 The monopoly quantity is $\frac{a-c}{2}$. The monopoly price is, therefore:

$$P = a - Q = q - \left(\frac{a-c}{2}\right) = \frac{a+c}{2}$$

The players can construct the following strategy.

- In Round 1, play $p_i = \frac{a+c}{2}$
- In Round t≠1, if the previous Round had $p_j \neq \frac{a+c}{2}$, play $p_i = c$ (the Bertrand equilibrium). Else, play $p_i = \frac{a+c}{2}$

Now, if player i deviates by charging $\frac{a-c}{2} - \varepsilon$ where $\varepsilon > 0$, he secures the entire market and earns monopoly profits for one round and Bertrand profits (which are 0) for all future rounds which totals to

$$\pi_{deviate} = \left(a - \frac{a-c}{2} - c\right) \cdot \left(\frac{a-c}{2}\right) + \delta \cdot 0 + \delta^2 \cdot 0 + \ldots = \frac{(a-c)^2}{4}$$

The payoff from sticking to the strategy (in which both firms produce $\frac{a-c}{4}$) the payoff is

$$\pi_{follow} = \left(a - \frac{a-c}{2} - c\right) \cdot \left(\frac{a-c}{4}\right) + \delta \cdot \left(a - \frac{a-c}{2} - c\right) \cdot \left(\frac{a-c}{4}\right) + \ldots$$

$$= \left(\frac{1}{1-\delta}\right) \cdot \left(\frac{a-c}{2}\right) \cdot \left(\frac{a-c}{4}\right) = \left(\frac{1}{1-\delta}\right) \cdot \frac{(a-c)^2}{8}$$

The strategy is stable if

$$\pi_{deviate} \leq \pi_{follow}$$

$$\Rightarrow \frac{(a-c)^2}{4} \leq \left(\frac{1}{1-\delta}\right) \cdot \frac{(a-c)^2}{2}$$

$$\Rightarrow \delta \geq \frac{1}{2}$$

Q.E.D.

Answer 2.14 The monopoly quantity when demand is high is $\frac{a_H - c}{2}$ which makes the monopoly price

$$p_H = a_H - \left(\frac{a_H - c}{2}\right) = \frac{a_H + c}{2}$$

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This is the price that the firms have to maintain when demand is high. Conversely, when demand is low

$$p_L = \frac{a_L + c}{2}$$

Let p_M be the monopoly price, defined as

$$p_M = \begin{cases} p_H & \text{if } a_i = a_H \\ p_L & \text{if } a_i = a_L \end{cases}$$

Consider the following strategy for firm i:

- In Round 1, set $p_i = p_M$.
- In Round $t \neq 1$, if $p_j = p_M$ in the previous round, play p_M, else play $p_i = c$

The payoff received from deviating is the monopoly profit for one round [4] and then zero profits in all future rounds:

[4] See the previous question for the derivation of one round monopoly profits

$$\pi_{deviate} = \frac{(a_i - c)^2}{4} + \delta \cdot 0 + \delta^2 \cdot 0 + \ldots = \frac{(a_i - c)^2}{4}$$

If he follows the strategy, he earns:

$$\pi_{follow} = \frac{(a_i - c)^2}{8} + \delta \cdot \left(\pi \cdot \frac{(a_H - c)^2}{8} + (1 - \pi) \cdot \frac{(a_L - c)^2}{8} \right) + \ldots$$

$$\Rightarrow \pi_{follow} = \frac{(a_i - c)^2}{8} + \frac{\delta}{1 - \delta} \cdot \left(\pi \cdot \frac{(a_H - c)^2}{8} + (1 - \pi) \cdot \frac{(a_L - c)^2}{8} \right)$$

The strategy is stable if

$$\pi_{deviate} \leq \pi_{follow}$$

$$\Rightarrow \frac{(a_i - c)^2}{4} \leq \frac{(a_i - c)^2}{8} + \frac{\delta}{1 - \delta} \cdot \left(\pi \cdot \frac{(a_H - c)^2}{8} + (1 - \pi) \cdot \frac{(a_L - c)^2}{8} \right)$$

Answer 2.15 If the quantity produced by a monopolist is $\frac{a-c}{2}$, the quantity produced by a single company in a successful cartel is

$$q_n^m = \frac{a - c}{2n}$$

Therefore, the profit earned by one of these companies is

$$\pi_m = (p - c)q_n^m = (a - Q - c)q_n^m = \left(a - \frac{a - c}{2} - c \right) \left(\frac{a - c}{2n} \right) = \frac{1}{n} \cdot \left(\frac{a - c}{2} \right)^2$$

The Cournot oligopoly equilibrium quantity[5] is $\frac{a-c}{1+n}$ which means that the profit earned at this equilibrium is

[5] See Answer 1.4

$$\pi_c = (a - Q - c)q_n^c = \left(a - n \left[\frac{a - c}{1 + n} \right] - c \right) \cdot \left(\frac{a - c}{1 + n} \right) = \left(\frac{a - c}{1 + n} \right)^2$$

A grim trigger strategy for a single company here is

- In Round t=1, produce q_n^m

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• In Round $t > 1$, if the total quantity produced in $t-1$ is $n \cdot q_n^m$, produce q_n^m, else produce q_n^c.

Now, the best response to everyone else producing q_n^m is determined by finding q' which renders profit

$$\pi' = (a - Q - c)q' = \left(a - \frac{a-c}{2n} \cdot (n-1) - q' - c\right) q' = \frac{n+1}{2n} \cdot (a-c)q' - q'^2$$

Which is maximized at

$$\frac{d\pi'}{dq'} = \frac{n+1}{2n} \cdot (a-c) - 2q' = 0 \Rightarrow q' = \frac{a+1}{4n}(a-c)$$

The profit at q' (cheating gain) is

$$\pi' = \left(a - c - \frac{a-c}{2n} \cdot (n-1) - \frac{n+1}{4n} \cdot (a-c)\right) \cdot \left(\frac{n+1}{4n} \cdot (a-c)\right) = \left[\frac{n+1}{4n} \cdot (a-c)\right]^2$$

If the firm deviates, it earns the cheating gain for one round and Cournot profits for all future rounds i.e. the gain from deviating from the strategy is

$$\pi_{deviate} = \left(\frac{n+1}{4n} \cdot (a-c)\right)^2 + \delta \left(\frac{a-c}{1+n}\right)^2 + \delta^2 \left(\frac{a-c}{1+n}\right)^2 + \dots$$

$$\Rightarrow \pi_{deviate} = \left[\left(\frac{n+1}{4n}\right)^2 + \frac{\delta}{1+\delta}\left(\frac{1}{1+n}\right)^2\right](a-c)^2$$

If the firm follows the strategy, it's payoff is π_m for all rounds:

$$\pi_{follow} = \frac{1}{n}\left(\frac{a-c}{2}\right)^2 + \delta \cdot \frac{1}{n}\left(\frac{a-c}{2}\right)^2 + \delta^2 \cdot \frac{1}{n}\left(\frac{a-c}{2}\right)^2 + \dots = \frac{1}{1-\delta} \cdot \frac{1}{n}\left(\frac{a-c}{2}\right)^2$$

The strategy is stable if

$$\pi_{follow} \geq \pi_{deviate}$$

$$\Rightarrow \frac{1}{1-\delta} \cdot \frac{1}{n}\left(\frac{a-c}{2}\right)^2 \geq \left[\left(\frac{n+1}{4n}\right)^2 + \frac{\delta}{1-\delta}\left(\frac{1}{n+1}\right)^2\right](a-c)^2$$

$$\Rightarrow \delta^* \leq \frac{n^2+2n+1}{n^2+6n+1}$$

Thus as n rises, δ^* falls. If you want to know more, see Rotember and Saloner (1986).

Answer 2.16

Answer 2.17

Answer 2.18 See text.

Answer 2.19 In a one period game, player 1 would get a payoff of 1 and player 2 would get a payoff of 0 i.e. (1,0). In a two period

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game, if the two players can't agree, the game goes to the second stage, at which point, player 2 gets 1 and player 1 gets 0. This payoff of 1 in the second round is worth δ to player 2 in the first round. If player 1 offers δ to player 2 in the first round, player 2 will accept, getting a payoff of $1-\delta$ i.e. $(1-\delta, \delta)$.

In a three period game, if player 2 rejects the offer in the first round, they go on to the second round, at which point it becomes a two period game and player 2 gets a payoff of $1-\delta$ and player 1 gets δ. This payoff $(\delta,1-\delta)$ is worth $(\delta^2,\delta[1-\delta])$ to the players in the first round. Thus, if player 1 makes an offer of $(1-\delta[1-\delta],\delta[1-\delta])$=$(1-\delta+\delta^2,\delta-\delta^2)$, player 2 would accept and player 1 would secure a higher payoff.

Answer 2.20 In round 1, player A offers $(\frac{1}{1-\delta}, \frac{\delta}{1+\delta})$ to player B, which player B accepts since $\frac{\delta}{1+\delta} \geq \delta s^* = \delta\frac{1}{1-\delta}$.

What if A deviates from the equilibrium, offers less and B refuses? The game then goes into the next round and B offers A $\frac{\delta}{1+\delta}$, which will be accepted, leaving $\frac{1}{1+\delta}$ for B. This is worth $\delta \cdot \frac{1}{1+\delta}$ to B in the first round (which is why B will refuse anything less than this amount) and $\frac{\delta^2}{1+\delta}$ to A in the first round. This is less than the $\frac{1}{1+\delta}$ A *would* have made had he not deviated.

Answer 2.21

Answer 2.22 In the first round, investors can either withdraw (w) or not (d). A strategy can represented as x_1x_2 where x_1 is what the investor does in the first round and x_2 is what the investor does in the second round. The game can be represented by the following table:

	ww	wd	dd	dw
ww	<u>r</u>,<u>r</u>	<u>r</u>,<u>r</u>	D,2r-D	D,2r-D
wd	<u>r</u>,<u>r</u>	<u>r</u>,<u>r</u>	D,2r-D	D,2r-D
dd	2r-D,D	2r-D,D	R,R	D,<u>2R-D</u>
dw	2r-D,D	2r-D,D	<u>2R-D</u>,D	<u>R</u>,<u>R</u>

There are 5 Nash Equilibria: (dw,dw) , (ww,ww) , (ww,wd) , (wd,ww) and (wd,wd). Of these, (ww,wd), (wd,ww) and (wd,wd) are not Subgame Perfect Equilibria since there is no subgame in which both or either player doesn't withdraw his or her funds in the second round.

Answer 2.23 The optimal investment is given by

$$\frac{d(v+I-p-I^2)}{dI} = 0 \Rightarrow I^* = \frac{1}{2}$$

The boost in the value added is If the buyer had played 'Invest', buyer will buy if

$$v+I-p-I^2 \geq -I^2 \Rightarrow p \leq v+I$$

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Thus, the highest possible price that the buyer will pay at this point is $p = v + I$. If, however, the buyer doesn't invest, the buyer will buy if

$$v - p \geq 0 \Rightarrow v \geq p$$

Thus the buyer would be willing to pay $p = v$. Thus investment is $I \subseteq \{0, \frac{1}{2}\}$. The price is drawn from $p \subseteq \{v, v + \frac{1}{2}\}$. There is no gain from charging anything other than these prices.

	$v + \frac{1}{2}$	v
$\frac{1}{2}, A$	$-\frac{1}{4}, \underline{v + \frac{1}{2}}$	$\underline{\frac{1}{4}}, v$
$\frac{1}{2}, R$	$-\frac{1}{4}, \underline{0}$	$-\frac{1}{4}, \underline{0}$
$0, A$	$-\frac{1}{2}, \underline{v + \frac{1}{2}}$	$0, v$
$0, R$	$\underline{0}, \underline{0}$	$0, \underline{0}$

As you can see from this (complicated) table, if $I = \frac{1}{2}$, A weakly dominates R. And if $I = 0$ R weakly dominates A. Thus we can collapse the above table into a simpler one:

	$(q)v + \frac{1}{2}$	$(1-q)v$
$(p)I = \frac{1}{2}_{Accept}$	$-\frac{1}{4}, \underline{v + \frac{1}{2}}$	$\underline{\frac{1}{4}}, v$
$(1-p)I = 0_{Reject}$	$\underline{0}, \underline{0}$	$0, \underline{0}$

The only pure Nash Equilibrium is for the buyer to not invest and the

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Static Games of Incomplete Information

Answer 3.1 See text

Answer 3.2 Firm 1 aims to maximize

$$\pi_1 = (p - c)q_1 = (a_i - c - q_1 - q_2)q_1$$

Which is done by

$$\frac{d\pi_1}{dq_1} = a_i - c - q_1 - q_2 + q_1(-1) = 0 \Rightarrow q_1 = \frac{a_i - c - q_2}{2}$$

Thus, the strategy for firm 1 is

$$q_1 = \begin{cases} \frac{a_H - c - q_2}{2} & \text{if } a_i = a_H \\ \frac{a_L - c - q_2}{2} & \text{if } a_i = a_L \end{cases}$$

Now, the firm 2 aims to maximize

$$\pi_2 = (p - c)q_2 = (a - c - q_1 - q_2)q_2$$

This is maximized at

$$\frac{d\pi_2}{dq_2} = a - c - q_1 - q_2 + q_2(-1) = 0 \Rightarrow q_2 = \frac{a - c - q_1}{2} = \frac{\theta a_H + (1-\theta)a_L - c - q_1}{2}$$

Plugging in q_1, we get

$$q_2 = \frac{\theta a_H + (1-\theta)a_L - c - \left[\frac{\theta a_H + (1-\theta)a_L - c - q_2}{2}\right]}{2}$$

$$\Rightarrow q_2 = \frac{\theta a_H + (1-\theta)a_L - c}{3}$$

Now, we need to find out what firm 1's output would be. If $a_i = a_H$,

$$q_1^H = \frac{a_H - c - \frac{\theta a_H + (1-\theta)a_L - c}{3}}{2} = \frac{(3-\theta)a_H - (1-\theta)a_L - 2c}{6}$$

But what if $a_i = a_L$?

$$q_1^L = \frac{a_L - c - \frac{\theta a_H + (1-\theta)a_L - c}{3}}{2} = \frac{(2+\theta)a_L - \theta a_H - 2c}{6}$$

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Now, based on these results, the constraints for non-negativity are:

$$q_2 \geq 0 \Rightarrow \theta a_H + (1-\theta)a_L - c \geq 0 \Rightarrow \theta \geq \frac{c - a_L}{a_H - a_L}$$

Which also requires

$$\theta \leq 1 \Rightarrow 1 \geq c - a_L \Rightarrow a_L \geq c - 1$$

Furthermore,

$$q_1^L \geq 0 \Rightarrow \frac{(2+\theta)a_L - \theta a_H - 2c \geq}{6} \geq 0 \Rightarrow \theta \leq 2 \cdot \frac{c - a_L}{a_H - a_L}$$

Which subsumes the last-but-one result. And finally,

$$q_1^H \geq 0 \Rightarrow \theta \leq \frac{2c - 3a_H + a_L}{a_H - a_L}$$

Answer 3.3 The profits earned by firm 1 is given by

$$\pi_1 = (p_1 - c)q_1 = (p_1 - c)(a - p_1 - b_1 p_2)$$

This is maximized at

$$\frac{d\pi_1}{dp_1} = a - p_1 - b_1 p_2 + p_1(-1) = 0 \Rightarrow p_1 = \frac{a - b_1 p_2}{2} = \frac{a - b_1[\theta p_H + (1-\theta)p_L]}{2}$$

Now, what if $b_1 = b_H$? To start with, $p_1 = p_H$ and

$$p_H = \frac{a - b_H[\theta p_H + (1-\theta)p_L]}{2} \Rightarrow p_H = \frac{a - (1-\theta)b_H p_L}{2 + \theta b_H}$$

And if $b_1 = b_L$:

$$p_L = \frac{a - b_L[\theta p_H + (1-\theta)p_L]}{2} = \frac{a - \theta b_L p_H}{2 + b_L(1-\theta)}$$

Which means that,

$$p_H = \frac{a - (1-\theta)\left[\frac{a - \theta b_L p_H}{2 + (1-\theta)b_L}\right]}{2 + \theta b_H}$$

$$\Rightarrow p_H = \frac{a(1 - [1-\theta]b_H)}{4 + 2(1-\theta)b_L + 2\theta b_H}$$

Similarly,

$$p_L = \frac{a(1 - \theta b_L)}{4 + 2(1-\theta)b_L + 2\theta b_H}$$

Answer 3.4 Game 1 is played with 0.5 probability:

	L	R
(q)T	1,1	0,0
(1-q)B	0,0	0,0

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If nature picks game 2, which 0.5 probability:

	L	R
(q)T	0,0	0,0
(1-q)B	0,0	2,2

If nature picks game 2, player 1 will always play B, since it weakly dominates T and player 2 will play R, since it weakly dominates L.

Now, if nature chooses game 1, player 1 will play T. If nature chooses game 2, player 1 will play B. Furthermore, if player 2 plays L with probability p:

$$\pi_2 = p[\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 1] + (1-p)[\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 2] = 1 - \frac{1}{2} \cdot p$$

This is maximized at $p = 0$ i.e. player 2 will always play R. Thus the Pure-strategy Bayesian Nash equilibrium is

$$PSNE = \{(1, T, R), (2, B, R)\}$$

Answer 3.5

Answer 3.6 The payoff is given by

$$u_i = \begin{cases} v_i - b_i & \text{if } b_i > b_j \ \forall j = 1, 2, \ldots, i-1, i+1, \ldots, n \\ \frac{v_i - b_j}{m} & \text{if } b_i = b_j \\ 0 & \text{if } b_i < b_j \text{ for any } j = 1, 2, \ldots, i-1, i+1, \ldots, n \end{cases}$$

The beliefs are: v_j is uniformly distributed on [0,1]. Actions are given by $b_i \subseteq [0, 1]$ and types are given by $v_i \subseteq [0, 1]$. The strategy is given by $b_i = a_i + c_i v_i$. Thus, the aim is to maximize

$$\pi_i = (v_i - b_i) \cdot \mathbb{P}(b_i > b_j \forall j = 1, \ldots, i-1, i+1, \ldots, n) = (v_i - b_i) \cdot [\mathbb{P}(b_i > b_j)]^{n-1}$$

$$\Rightarrow \pi_i = (v_i - b_i) \cdot [\mathbb{P}(b_i > a_j + c_j v_j)]^{n-1} = (v_i - b_i) \cdot \left[\mathbb{P}\left(v_j < \frac{b_i - a_j}{c_j}\right)\right]^{n-1} = (v_i - b_i)\left(\frac{b_i - a_j}{c_j}\right)^{n-1}$$

This is maximized at

$$\frac{d\pi_i}{db_i} = (-1) \cdot \left(\frac{b_i - a_j}{c_j}\right)^{n-1} + (v_i - b_i)\left(\frac{n-1}{c_j}\right)\left(\frac{b_i - a_j}{c_j}\right)^{n-2} = 0$$

$$\left(\frac{b_i - a_j}{c_j}\right)^{n-2} \cdot \left(\frac{a_j + (n-1)v_i - nb_i}{c_j}\right) = 0$$

This requires that either

$$\frac{b_i - a_j}{c_j} = 0 \Rightarrow b_i = a_j$$

Or that

$$\frac{a_j + (n-1)v_i - nb_i}{c_j} = 0 \Rightarrow b_i = \frac{a_j}{n} + \frac{n-1}{n} \cdot v_i$$

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Now, we know that $b_i = a_i + c_i v_i$. Here, we know that $c_i = \frac{n-1}{n}$ and

$$a_i = \frac{a_j}{n} \Rightarrow a_1 = \frac{a_2}{n} = \frac{a_3}{n} = \ldots = \frac{a_n}{n}$$

Which is only possible if $a_1 = a_2 = \ldots = a_n = 0$. Thus,

$$b_i = \frac{n-1}{n} \cdot v_i$$

Answer 3.7

Answer 3.8

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Dynamic Games of Incomplete Information

Answer 4.1 (a)

	(q)L′	(1-q)R′
L	$\underline{4}$,$\underline{1}$	0,0
M	3,0	0,$\underline{1}$
R	2,$\underline{2}$	$\underline{2}$,$\underline{2}$

The Nash Equilibria are $(L, L'), (R, R')$ and they are both sub-game perfect equilibria. Now, the payoff to player 2 from playing L' is

$$\pi_2(L') = 1 \cdot p + 0 \cdot (1 - p) = p$$

The payoff from playing R'

$$\pi_2(R') = 0 \cdot p + 1 \cdot (1 - p) = 1 - p$$

Player 2 will always play L' if

$$\pi_2(L') > \pi_2(R') \Rightarrow p > 1 - p \Rightarrow p > \frac{1}{2}$$

The playoff to player 1 from playing L is

$$\pi_1(L) = 4 \cdot q + 0 \cdot (1 - q) = 4q$$

And the payoff from playing M is

$$\pi_1(M) = 3 \cdot q + 0 \cdot (1 - q) = 3q$$

Player 1 will always play L if

$$\pi_1(L) > \pi_1(M) \Rightarrow 4q > 3q$$

Which is true. Thus $p = 1$. In which case, player 2 will always play L'. Thus the outcome (R, R') violates Requirements 1 and 2.

(b)

	L'	M'	R'
L	1,$\underline{3}$	1,2	$\underline{4}$,0
M	$\underline{4}$,0	0,2	3,$\underline{3}$
R	2,$\underline{4}$	$\underline{2}$,$\underline{4}$	2,$\underline{4}$

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The expected values of the payoffs to player 2 are:

$$\pi_2(L') = 3 \cdot p + 0 \cdot (1-p) = 3p$$

$$\pi_2(M') = 2 \cdot p + 2 \cdot (1-p) = 2$$

$$\pi_2(R') = 0 \cdot p + 3 \cdot (1-p) = 3 - 3p$$

And the payoffs to player 1 are:

$$\pi_1(R) = 2$$

$$\pi_1(L) = 1 \cdot q_1 + 1 \cdot q_2 + 4 \cdot (1 - q_1 - q_2) = 4 - 3q_1 - 3q_2$$

$$\pi_1(M) = 4 \cdot q_1 + 0 \cdot q_2 + 3 \cdot (1 - q_1 - q_2) = 3 + q_1 - 3q_2$$

The only Nash Equilibrium is (R, M'); it is also sub-game perfect.
To be a Perfect Bayesian Equilibrium, player 2 must believe that

$$\pi_2(M') > \pi_2(L') \Rightarrow 2 > 3p \Rightarrow \frac{2}{3} > p$$

and

$$\pi_2(M') > \pi_2(R') \Rightarrow 2 > 3 - 3p \Rightarrow p > \frac{1}{3}$$

Furthermore, player 1 must believe

$$\pi_1(R) > \pi_1(L) \Rightarrow 2 > 4 - 3q_1 - 3q_2 \Rightarrow q_1 > \frac{2}{3} - q_2$$

Since $q_1 > 0$, this implies that

$$\frac{2}{3} - q_2 > 0 \Rightarrow \frac{2}{3} > q_2$$

and

$$\pi_1(R) > \pi_1(M) \Rightarrow 2 > 3 + q_1 - 3q_2 \Rightarrow 3q_2 - 1 > q_1$$

Which, in turn, requires

$$3q_2 - 1 > \frac{2}{3} - q_2 \Rightarrow q_2 > \frac{5}{12}$$

The pure Bayesian Nash equilibrium is

$$\left[(R, M), \frac{1}{3} > p > \frac{2}{3}, 3q_2 - 1 > q_1 > \frac{2}{3} - q_2, \frac{2}{3} > q_2 > \frac{5}{12}\right]$$

Answer 4.2

	$(q)L'$	$(1-q)R'$
$(p)L$	$\underline{3}$,0	0,$\underline{1}$
$(1-p)M$	0,$\underline{1}$	$\underline{3}$,0
R	2,$\underline{2}$	2,$\underline{2}$

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As you can see, there is no Pure Nash Equilibrium. But, we need rigorous proof: A pure strategy Nash Equilibrium exists if

(a) Player 1 always picks either L or M. For example, player 1 will always play L if

$$\pi_1(L) > \pi_1(M) \Rightarrow 3 \cdot q + 0 \cdot (1-q) > 0 \cdot q + 3 \cdot (1-q) \Rightarrow q > \frac{1}{2}$$

Thus if $q > 0.5$, $p = 1$.

(b) Player 2 always picks either L' or R'; player 2 will always play L' if

$$\pi_2(L') > \pi_2(R') \Rightarrow 0 \cdot p + 1 \cdot (1-p) > 1 \cdot p + 0 \cdot (1-p) \Rightarrow \frac{1}{2} > p$$

Thus, if $p < 0.5$, $q = 1$. This violates the condition we uncovered in part (a), proving that there is no PSNE.

In a mixed strategy BE, player 1 plays L with probability p and player 2 plays L' with probability q. In equilibrium, player 2 is indifferent between L' and R':

$$\pi_2(L') = \pi_2(R') \Rightarrow 0 \cdot p + 1 \cdot (1-p) = 1 \cdot p + 0 \cdot (1-p) \Rightarrow p = \frac{1}{2}$$

And similarly, for player 1:

$$\pi_1(L) = \pi_1(M) \Rightarrow 3 \cdot q + 0 \cdot (1-q) = 0 \cdot q + 3 \cdot (1-q) \Rightarrow q = \frac{1}{2}$$

Thus, in a mixed strategy equilibrium, player 1 plays R with $p = 0.5$ and player 2 plays L' with probability $q = 0.5$.

Answer 4.3 (a) Let's start with the pooling equilibrium (R, R). In this situation, $p = 0.5$. Now, the payoff to the receiver is

$$\pi_R(R, u) = 0.5 \cdot (1) + 0.5 \cdot (0) = 0.5$$

$$\pi_R(R, d) = 0.5 \cdot (0) + 0.5 \cdot (2) = 1$$

Thus, if the sender plays R, the receiver will play d. We have to test two strategies for the receiver: (u, d) and(d, d). Under the strategy (d, d)

$$\pi_1(L, d) = 2 \text{ and } \pi_1(R, d) = 3$$

$$\pi_2(L, d) = 3 \text{ and } \pi_2(R, d) = 2$$

There is no incentive for type 1 to deviate and play L, but there is an incentive for type 2 to do so. Under the strategy (u, d),

$$\pi_1(L, u) = 1 \text{ and } \pi_1(R, d) = 3$$

$$\pi_2(L, u) = 0 \text{ and } \pi_2(R, d) = 2$$

Neither type 1 nor type 2 have any reason to L instead of R. Thus, we have the following pooling equilibrium:

$$[(R, R), (u, d), p = 0.5, 1 \geq q \geq 0]$$

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(b) We must find a pooling equilibrium in which the sender plays (L, L, L). For the receiver, the payoffs are

$$\pi_R(L, u) = \frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 1 = 1$$

$$\pi_R(L, d) = \frac{1}{3} \cdot 0 + \frac{1}{3} \cdot 0 + \frac{1}{3} \cdot 0 = 0$$

There are two strategies: (u, u) and (u, d). Under (u, u):

$$\pi_1(L, u) = 1 \text{ and } \pi_1(R, u) = 0$$

$$\pi_2(L, u) = 2 \text{ and } \pi_2(R, u) = 1$$

$$\pi_3(L, u) = 1 \text{ and } \pi_3(R, u) = 0$$

None of the three types have an incentive to send R instead of L. Thus, we have the following equilibrium:

$$[(L, L, L), (u, u), p = \frac{1}{3}, 1 \geq q \geq 0]$$

Answer 4.4 (a) Let's examine pooling equilibrium (L, L). $p = 0.5$. $\pi_R(L, u) = \pi_R(L, d)$. Thus, it doesn't matter for the receiver whether he/she plays u or d.

• Under (u, u), $\pi_1(L, u) = 1 < \pi_1(R, u) = 2$, making it unsustainable.

• Under (u, d), $\pi_2(L, u) = 0 < \pi_2(R, d) = 1$, making it unsustainable.

• Under (d, d), $\pi_2(L, d) = 0 < \pi_2(R, d) = 1$, making it unsustainable.

• Under (d, u), $\pi_2(L, u) = 0 < \pi_2(R, d) = 1$, making it unsustainable.

Thus, (L, L) is not a sustainable equilibrium.

Let's examine separating equilibrium (L, R). The best response to this is (u, d)[6]. Let's see if either of the types have an incentive to deviate:

• For type 1, $\pi_1(L, u) = 1 > \pi_1(R, d) = 0$ i.e. no reason to play R instead of L.

• For type 2, $\pi_2(L, u) = 0 < \pi_2(R, d) = 1$ i.e. no reason to play L instead of R.

Let's examine pooling equilibrium (R, R). $\pi_R(R, u) = 1 > \pi_R(R, d) = 0.5$. Therefore, the two strategies that can be followed by the receiver are (u, u) and (d, u).

• Under (u, u), $\pi(L, u) < \pi(R, u)$ for both types.

• Under (d, u), $\pi(L, d) \leq \pi(R, u)$ for both types.

Let's examine pooling equilibrium (R, L). The best response to this is (d, u)[7].

• For type 1, $\pi_1(L, d) = 2 \geq \pi_1(R, u) = 2$, i.e. will play L.

• For type 2, $\pi_2(L, d) = 0 \leq \pi_2(R, u) = 1$, i.e. will play R.

[6] Since $\pi_R(1, L, u) > \pi_R(1, L, d)$ and $\pi_R(2, R, u) > \pi_R(2, R, d)$

[7] Since $\pi_R(1, R, u) = 2 > \pi_R(1, R, d) = 0$ and $\pi_R(2, L, u) = 0 < \pi_R(2, L, d) = 1$

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Thus the perfect Bayesian equilibrium are:

$$[(L,R),(u,d),p,q]$$

$$[(R,R),(u,u),p,q=0.5]$$

$$[(R,R),(d,u),p,q=0.5]$$

$$[(R,L),(d,u),p,q]$$

(b) Let's examine pooling equilibrium is (L,L). $\pi_R(L,u)=1.5>\pi_R(L,d)=1$, therefore player 2 will respond to L with u. The two strategies are (u,u) and (u,d).

• Under (u,u), $\pi(L,u)>\pi(R,u)$ for both types.

• Under (u,d), $\pi_1(L,u)=3<\pi_1(R,d)=4$, making it unsustainable.

Let's examine separating equilibrium (L,R). The best response to this is (d,u).

• For type 1, $\pi_1(L,d)=1>\pi_1(R,u)=0$ i.e. type 1 will play L.

• For type 2, $\pi_2(L,d)=0<\pi_2(R,u)=1$ i.e. type 2 will play R.

Let's examine pooling equilibrium (R,R). $\pi_R(R,u)=1>\pi_R(L,d)=0.5$, therefore player 2 will respond to R with u. The two strategies are (u,u) and (d,u).

• Under (u,u), $\pi_1(L,u)=3>\pi_1(R,u)=0$, making (R,R) unsustainable.

• Under (d,u), $\pi_1(L,d)=1>\pi_1(R,u)=0$, making (R,R) unsustainable.

Let's examine separating equilibrium (R,L). The best response to this is (d,u).

• For type 1, $\pi_1(L,d)=1>\pi_1(R,u)=0$, making (R,L) unsustainable.

• For type 2, $\pi_2(L,d)=0<\pi_2(R,u)=1$, which doesn't conflict with the equilibrium.

The perfect Bayesian Equilibria are

$$[(L,L),(u,u),p=0.5,q]$$

$$[(L,R),(d,u),p=1,q=0]$$

Answer 4.5 Let's examine 4.3(a). We've already tested equilibrium (R,R). Let's try another pooling equilibrium (L,L). $q=0.5$. $\pi_R(L,u)=1>\pi_R(L,d)=0.5$. Thus, the receiver's response to L will always be u. We have to test two strategies for the receiver: (d,u) and(u,u).

• Under the strategy (d,u), $\pi(L,d)=2>\pi(R,u)=0$ i.e. there is no incentive for either type to play R instead of L.

• Under the strategy (u,u), $\pi_2(L,u)=0<\pi_2(R,u)=1$, making (L,L) unsustainable.

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Let's examine (L, R). The best response to this is (u, d). In response to this,

• For type 1, $\pi_1(L, u) = 1 < \pi_1(R, d) = 3$ i.e. type 1 will play R which violates the equilibrium

• For type 2, $\pi_2(L, u) = 0 < \pi_2(R, d) = 2$ i.e. type 2 will play R which doesn't violate the equilibrium.

Let's examine (R, L). The best response to this is (u, d). In response to this,

• For type 1, $\pi_1(L, u) = 1 < \pi_1(R, d) = 3$, i.e. type 1 will play R.

• For type 2, $\pi_2(L, u) = 0 < \pi_2(R, d) = 2$, i.e. type 2 will play R, violating the equilibrium.

The perfect Bayesian Equilibrium is

$$[(L, L), (d, u), p, q = 0.5]$$

Now, let's examine 4.3(b). There is one pooling equilibrium other than the (L, L, L): (R, R, R). There are six pooling equilibria: 1.(L, L, R), 2.(L, R, L), 3.(R, L, L), 4.(L, R, R), 5.(R, L, R) and 6.(R, R, L).

Let's start with pooling equilibrium (R, R, R). $\pi_R(R, u) = \frac{2}{3} > \pi_R(R, d) = \frac{1}{3}$. Thus, receiver will play the strategy (u, u) or (d, u).

• For strategy (u, u), $\pi(L, u) < \pi(R, u)$ for all types, making it unsustainable.

• For strategy (d, u), $\pi_1(L, d) < \pi_1(R, u)$, making the equilibrium unsustainable. Let's examine the various separating the equilibrium.

1. (L, L, R). The best response to this is (u, d), [8]

• For type 1, $\pi_S(1, L, u) = 1 > \pi_S(1, R, d) = 0$ i.e. type 1 will play L.

• For type 2, $\pi_S(2, L, u) = 2 > \pi_S(2, R, d) = 1$ i.e. type 2 will play L.

• For type 3, $\pi_S(3, L, u) = 1 < \pi_S(3, R, d) = 2$ i.e. type 3 will play R.

Thus, this is a viable equilibrium.

[8] Since $\pi_R(3, R, u) = 0 < \pi_R(3, R, d) = 1$ and $0.5 \cdot \pi_R(1, L, u) + 0.5 \cdot \pi_R(2, L, u) = 1 > 0.5 \cdot \pi_R(1, L, d) + 0.5 \cdot \pi_R(1, L, d) = 0$

2. (L, R, L). The best response to this is (u, u), [9]

• For type 1, $\pi_S(1, L, u) = 1 > \pi_S(1, R, u) = 0$ i.e. type 1 will play L.

• For type 2, $\pi_S(2, L, u) = 2 > \pi_S(2, R, u) = 1$ i.e. type 2 will play L, instead of R.

• For type 3, $\pi_S(3, L, u) = 1 > \pi_S(3, R, u) = 0$ i.e. type 3 will play L.

Thus, this is not a viable equilibrium.

[9] since $\pi_R(2, R, u) = 1 > \pi_R(2, R, d) = 0$ and $0.5 \cdot \pi_R(1, L, u) + 0.5 \cdot \pi_R(3, L, u) = 1 > 0.5 \cdot \pi_R(1, L, d) + 0.5 \cdot \pi_R(3, L, d) = 0$

3. (R, L, L). The best response to this is (u, u), [10]

• For type 1, $\pi_S(1, L, u) = 1 > \pi_S(1, R, u) = 0$ i.e. type 1 will play L, instead of R.

• For type 2, $\pi_S(2, L, u) = 2 > \pi_S(2, R, u) = 1$ i.e. type 2 will play L.

• For type 3, $\pi_S(3, L, u) = 1 > \pi_S(3, R, u) = 0$ i.e. type 3 will play L.

Thus, this is a not viable equilibrium.

[10] since $\pi_R(1, R, u) = 1 < \pi_R(1, R, d) = 0$ and $0.5 \cdot \pi_R(2, L, u) + 0.5 \cdot \pi_R(3, L, u) = 1 > 0.5 \cdot \pi_R(2, L, d) + 0.5 \cdot \pi_R(3, L, d) = 0$

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4. (L, R, R). The best response to this is (u, u) and (u, d) [11]
Let's test (u, u):
• For type 1, $\pi_S(1, L, u) = 1 > \pi_S(1, R, u) = 0$ i.e. type 1 will play L.
• For type 2, $\pi_S(2, L, u) = 2 > \pi_S(2, R, u) = 1$ i.e. type 2 will play L, instead of R.
• For type 3, $\pi_S(3, L, u) = 1 > \pi_S(3, R, u) = 0$ i.e. type 3 will play L, instead of R.
Thus, this is not a viable equilibrium.
Let's test (u, d)
• For type 1, $\pi_S(1, L, u) = 1 > \pi_S(1, R, d) = 0$ i.e. type 1 will play L.
• For type 2, $\pi_S(2, L, u) = 2 > \pi_S(2, R, d) = 1$ i.e. type 2 will play L, instead of R.
• For type 3, $\pi_S(3, L, u) = 1 < \pi_S(3, R, d) = 2$ i.e. type 3 will play R.
Thus, this is not a viable equilibrium.

[11] since $\pi_R(1, L, u) = 1 > \pi_R(1, L, d) = 0$ and $0.5 \cdot \pi_R(2, R, u) + 0.5 \cdot \pi_R(3, R, u) = 0.5 = 0.5 \cdot \pi_R(2, R, d) + 0.5 \cdot \pi_R(3, R, d) = 0.5$

5. (R, L, R). The best response to this is either (u, u) or (u, d) [12]
Let's test (u, u).
• For type 1, $\pi_S(1, L, u) = 1 > \pi_S(1, R, u) = 0$ i.e. type 1 will play L, instead of R.
• For type 2, $\pi_S(2, L, u) = 2 > \pi_S(2, R, u) = 1$ i.e. type 2 will play L.
• For type 3, $\pi_S(3, L, u) = 1 > \pi_S(3, R, u) = 0$ i.e. type 3 will play L, instead of R.
Thus, this is not a viable equilibrium.
Let's test (u, d)
• For type 1, $\pi_S(1, L, u) = 1 > \pi_S(1, R, d) = 0$ i.e. type 1 will play L, instead of R.
• For type 2, $\pi_S(2, L, u) = 2 > \pi_S(2, R, d) = 0$ i.e. type 2 will play L.
• For type 3, $\pi_S(3, L, u) = 1 = \pi_S(3, R, d) = 1$ i.e. type 3 can play R.
Thus, this is not a viable equilibrium.

[12] since $\pi_R(2, L, u) = 1 > \pi_R(2, L, d) = 0$ and $0.5 \cdot \pi_R(2, L, u) + 0.5 \cdot \pi_R(3, L, u) = 0.5 = 0.5 \cdot \pi_R(2, L, d) + 0.5 \cdot \pi_R(3, L, d) = 0.5$

6. (R, R, L). The best response to this is (u, u), [13]
• For type 1, $\pi_S(1, L, u) = 1 > \pi_S(1, R, u) = 0$ i.e. type 1 will play L, instead of R
• For type 2, $\pi_S(2, L, u) = 2 > \pi_S(2, R, u) = 1$ i.e. type 2 will play L, instead of R
• For type 3, $\pi_S(3, L, u) = 1 > \pi_S(3, R, u) = 0$ i.e. type 3 will play L.
Thus, this is not a viable equilibrium.
The only other perfect Bayesian Equilibrium is

[13] since $\pi_R(3, L, u) = 1 > \pi_R(3, L, d) = 0$ and $0.5 \cdot \pi_R(1, R, u) + 0.5 \cdot \pi_R(2, R, u) = 0.5 > 0.5 \cdot \pi_R(1, R, d) + 0.5 \cdot \pi_R(2, R, d) = 0$

$$\left[(L, L, R), (u, d), p, q_0 = \frac{1}{3}, q_1 = \frac{1}{3}\right]$$

Answer 4.6 Type 2 will always play R since $\pi_S(2, R, a) > \pi_S(2, R, u)$ and $\pi_S(2, R, a) > \pi_S(2, R, d)$. Thus if the Receiver gets the message L, he knows that it can only be type 1. In such a case, the Receiver plays u[14], creating

[14] in fact, $\pi_R(x, L, u) > \pi_R(x, L, d)$ for both types, so the Receiver will always play u

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a payoff of $(2, 1)$. This gives type 1 a higher payoff than if he played R, which would have given him a payoff of 1. Thus, the perfect Bayesian Equilibrium is

$$[(L, R), (u, a), p = 1, q = 0]$$

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